Chapter 7. The Zeta Function and Prime Number Theorem

Zeros of ζ on $\operatorname{Re} s = 1$, Chebyshev functions, proof of the prime number theorem

Contents

Overview

Euler's product formula for ζ connects analysis with the arithmetic of primes; an easy consequence is that $\sum_p 1/p$ diverges. To understand the distribution of primes, let

$$\pi(x) = \text{number of primes less than or equal to } x.$$

Legendre and Gauss conjectured

$$\pi(x) \sim \frac{x}{\log x} \qquad \text{as } x \to \infty, \tag{1}$$

where $f(x) \sim g(x)$ means $f(x)/g(x) \to 1$. Tchebychev proved by elementary methods the weaker $\pi(x) \approx x/\log x$, i.e. $A\frac{x}{\log x} \le \pi(x) \le B\frac{x}{\log x}$ for large x (2). In 1896 Hadamard and de la Vallée Poussin proved (1), the *prime number theorem*, using complex analysis, as we do. At the heart of the proof lies the fact that $\zeta(s)$ does not vanish on the line $\mathrm{Re}(s) = 1$ (the two statements are in fact equivalent).

1 Zeros of the zeta function

Motivation. The prime number theorem $\pi(x) \sim x/\log x$ is equivalent to the fact that ζ has no zeros on the line $\mathrm{Re}\, s = 1$. We prove this nonvanishing and the estimates for $1/\zeta$ needed to carry out the contour integration.

Euler's identity. For $\mathrm{Re}(s) > 1$,

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

Proof. Each factor is a convergent geometric series $1 + p^{-s} + p^{-2s} + \cdots$; formally multiplying them gives the result. Precisely, take first $s > 1$ real and integers $M > N$. By unique factorization, every $n \le N$ is a product of primes $\le N$, each repeated fewer than M times, so

$$\sum_{n=1}^{N} \frac{1}{n^s} \le \prod_{p \le N} \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots + \frac{1}{p^{Ms}}\right) \le \prod_{p \le N} \frac{1}{1 - p^{-s}} \le \prod_p \frac{1}{1 - p^{-s}},$$

and letting $N \to \infty$, $\sum_{n=1}^{\infty} n^{-s} \le \prod_p (1 - p^{-s})^{-1}$. Conversely, again by unique factorization, $\prod_{p \le N}(1 + p^{-s} + \cdots + p^{-Ms}) \le \sum_{n=1}^{\infty} n^{-s}$; letting $M \to \infty$, then $N \to \infty$, gives the reverse inequality. (For complex s with $\mathrm{Re}(s) > 1$ the identity follows by analytic continuation, or by the same argument with absolute convergence.) ∎

By Proposition 3.1, $\zeta(s) \neq 0$ for $\mathrm{Re}(s) > 1$. The functional equation $\xi(s) = \xi(1 - s)$ reads $\pi^{-s/2}\Gamma(s/2)\zeta(s) = \pi^{-(1-s)/2}\Gamma((1-s)/2)\zeta(1-s)$, so

$$\zeta(s) = \pi^{s-1/2}\frac{\Gamma((1-s)/2)}{\Gamma(s/2)}\zeta(1-s).$$

For $\mathrm{Re}(s) < 0$:

(i) $\zeta(1 - s)$ has no zeros, as $\mathrm{Re}(1 - s) > 1$;

(ii) $\Gamma((1 - s)/2)$ is zero free;

(iii) $1/\Gamma(s/2)$ has zeros at $s = -2, -4, -6, \ldots$. Hence:

Theorem 1.1

The only zeros of ζ outside the strip $0 \le \operatorname{Re}(s) \le 1$ are at the negative even integers, $-2, -4, -6, \ldots$.

The region $0 \le \operatorname{Re}(s) \le 1$ is the **critical strip**; the zeros outside it are the **trivial zeros**. Riemann conjectured the **Riemann hypothesis**: *the zeros of $\zeta(s)$ in the critical strip lie on the line* $\operatorname{Re}(s) = 1/2$—still one of the most famous unsolved problems. We prove:

Theorem 1.2

The zeta function has no zeros on the line $\operatorname{Re}(s) = 1$.

(Since ζ has a pole at 1 there are no zeros near $s = 1$; the point is the deeper fact $\zeta(1 + it) \neq 0$ for all $t \in \mathbb{R}$.) By the functional equation, ζ then has no zeros on $\operatorname{Re}(s) = 0$ either.

Lemma 1.3

If $\operatorname{Re}(s) > 1$, then

$$\log \zeta(s) = \sum_{p,m} \frac{p^{-ms}}{m} = \sum_{n=1}^{\infty} c_n n^{-s}$$

for some $c_n \ge 0$.

Proof. For $s > 1$, take the logarithm of Euler's product and use $\log \frac{1}{1-x} = \sum_{m=1}^{\infty} \frac{x^m}{m}$ $(0 \le x < 1)$:

$$\log \zeta(s) = \sum_{p} \log\Big(\frac{1}{1 - p^{-s}}\Big) = \sum_{p,m} \frac{p^{-ms}}{m}.$$

The double sum converges absolutely, so the order of summation is irrelevant (see the Note at the end of the chapter). The formula holds for $\operatorname{Re}(s) > 1$ by analytic continuation; $\log \zeta$ is well defined in the simply connected half-plane $\operatorname{Re}(s) > 1$ since ζ has no zeros there (Theorem 6.2). Finally $c_n = 1/m$ if $n = p^m$ and $c_n = 0$ otherwise. ∎

Lemma 1.4

If $\theta \in \mathbb{R}$, then $3 + 4\cos\theta + \cos 2\theta \ge 0$.

This follows from $3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2$.

Corollary 1.5

If $\sigma > 1$ and t is real, then $\log|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| \ge 0$.

Proof. With $s = \sigma + it$, $\operatorname{Re}(n^{-s}) = \operatorname{Re}(e^{-(\sigma+it)\log n}) = n^{-\sigma}\cos(t \log n)$. Hence, with $\theta_n = t \log n$,

$$\begin{aligned}\log|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| &= 3\operatorname{Re}[\log\zeta(\sigma)] + 4\operatorname{Re}[\log\zeta(\sigma + it)] + \operatorname{Re}[\log\zeta(\sigma + 2it)] \\ &= \sum c_n n^{-\sigma}(3 + 4\cos\theta_n + \cos 2\theta_n) \ge 0,\end{aligned}$$

by Lemma 1.4 and $c_n \ge 0$. ∎

Proof of Theorem 1.2. Suppose $\zeta(1 + it_0) = 0$ for some $t_0 \neq 0$. ζ is holomorphic at $1 + it_0$ and vanishes there to order ≥ 1, so $|\zeta(\sigma + it_0)|^4 \le C(\sigma - 1)^4$ as $\sigma \to 1$. Since $s = 1$ is a simple pole, $|\zeta(\sigma)|^3 \le C'(\sigma - 1)^{-3}$ as $\sigma \to 1$. And $|\zeta(\sigma + 2it_0)|$ stays bounded as $\sigma \to 1$. Hence $|\zeta^3(\sigma)\zeta^4(\sigma + it_0)\zeta(\sigma + 2it_0)| \to 0$ as $\sigma \to 1^+$, contradicting Corollary 1.5 (the logarithm of a number in $(0, 1)$ is negative). ∎

1.1 Estimates for $1/\zeta(s)$

The proof uses the logarithmic derivative ζ'/ζ near $\mathrm{Re}(s) = 1$, so we need growth estimates for ζ' (Proposition 2.7) and for $1/\zeta$. The following is a quantitative version of Theorem 1.2.

Proposition 1.6

For every $\epsilon > 0$, we have $1/|\zeta(s)| \le c_\epsilon |t|^\epsilon$ when $s = \sigma + it$, $\sigma \ge 1$, and $|t| \ge 1$.

Proof. From Corollary 1.5, $|\zeta^3(\sigma)\zeta^4(\sigma+it)\zeta(\sigma+2it)| \ge 1$ for $\sigma \ge 1$. With the estimate for ζ in Proposition 2.7 and $|\zeta(\sigma)| \le c/(\sigma-1)$,

$$|\zeta^4(\sigma+it)| \ge c|\zeta^{-3}(\sigma)||t|^{-\epsilon} \ge c'(\sigma-1)^3|t|^{-\epsilon},$$

so

$$|\zeta(\sigma+it)| \ge c'(\sigma-1)^{3/4}|t|^{-\epsilon/4} \qquad (\sigma \ge 1,\ |t| \ge 1). \tag{3}$$

Two cases, with a constant A to be chosen. If $\sigma - 1 \ge A|t|^{-5\epsilon}$, (3) gives $|\zeta(\sigma+it)| \ge A'|t|^{-4\epsilon}$, and replacing 4ϵ by ϵ gives the estimate. If $\sigma - 1 < A|t|^{-5\epsilon}$, choose $\sigma' > \sigma$ with $\sigma' - 1 = A|t|^{-5\epsilon}$. Then $|\zeta(\sigma+it)| \ge |\zeta(\sigma'+it)| - |\zeta(\sigma'+it) - \zeta(\sigma+it)|$, and by the mean value theorem and the bound on ζ' (Proposition 2.7), $|\zeta(\sigma'+it) - \zeta(\sigma+it)| \le c''|\sigma' - \sigma||t|^\epsilon \le c''(\sigma'-1)|t|^\epsilon$. With (3) at σ',

$$|\zeta(\sigma+it)| \ge c'(\sigma'-1)^{3/4}|t|^{-\epsilon/4} - c''(\sigma'-1)|t|^\epsilon.$$

Choose $A = (c'/(2c''))^4$; since $\sigma' - 1 = A|t|^{-5\epsilon}$, this makes $c'(\sigma'-1)^{3/4}|t|^{-\epsilon/4} = 2c''(\sigma'-1)|t|^\epsilon$, hence $|\zeta(\sigma+it)| \ge c''(\sigma'-1)|t|^\epsilon = A''|t|^{-4\epsilon}$. Replace 4ϵ by ϵ. ∎

2 Reduction to the functions ψ and ψ_1

Tchebychev introduced a function easier to handle than $\pi(x)$ but carrying the same information:

$$\psi(x) = \sum_{p^m \le x} \log p,$$

summed over integers of the form $p^m \le x$ (p prime, $m \ge 1$). Equivalently, with $\Lambda(n) = \log p$ if $n = p^m$ for some prime p and $m \ge 1$, and $\Lambda(n) = 0$ otherwise,

$$\psi(x) = \sum_{1 \le n \le x} \Lambda(n), \qquad \psi(x) = \sum_{p \le x} \left[\frac{\log x}{\log p}\right] \log p,$$

$[u]$ the greatest integer $\le u$ (since $p^m \le x$ iff $m \le \log x/\log p$).

Proposition 2.1

If $\psi(x) \sim x$ as $x \to \infty$, then $\pi(x) \sim x/\log x$ as $x \to \infty$.

Proof. We show $1 \le \liminf \pi(x)\frac{\log x}{x}$ and $\limsup \pi(x)\frac{\log x}{x} \le 1$ (4). Crudely, $\psi(x) = \sum_{p\le x}\left[\frac{\log x}{\log p}\right]\log p \le \sum_{p\le x}\frac{\log x}{\log p}\log p = \pi(x)\log x$, so $\frac{\psi(x)}{x} \le \frac{\pi(x)\log x}{x}$, and $\psi(x) \sim x$ gives the first inequality. For the second, fix $0 < \alpha < 1$:

$$\psi(x) \ge \sum_{p \le x} \log p \ge \sum_{x^\alpha < p \le x} \log p \ge (\pi(x) - \pi(x^\alpha))\alpha \log x,$$

so $\psi(x) + \alpha\pi(x^\alpha)\log x \ge \alpha\pi(x)\log x$. Dividing by x, and using $\pi(x^\alpha) \le x^\alpha$, $\alpha < 1$, $\psi(x) \sim x$, we get $1 \ge \alpha \limsup \pi(x)\frac{\log x}{x}$. As $\alpha < 1$ is arbitrary, the proof is complete. ∎

(The converse also holds.) It is more convenient to work with

$$\psi_1(x) = \int_1^x \psi(u)\,du.$$

Proposition 2.2

If $\psi_1(x) \sim x^2/2$ as $x \to \infty$, then $\psi(x) \sim x$ as $x \to \infty$, and therefore $\pi(x) \sim x/\log x$ as $x \to \infty$.

Proof. By Proposition 2.1 it suffices to show $\psi(x) \sim x$. Since ψ is increasing, for $\alpha < 1 < \beta$,

$$\frac{1}{(1-\alpha)x}\int_{\alpha x}^{x} \psi(u)\,du \le \psi(x) \le \frac{1}{(\beta-1)x}\int_{x}^{\beta x} \psi(u)\,du.$$

So $\psi(x) \le \frac{1}{(\beta-1)x}[\psi_1(\beta x) - \psi_1(x)]$, i.e. $\frac{\psi(x)}{x} \le \frac{1}{\beta-1}\left[\frac{\psi_1(\beta x)}{(\beta x)^2}\beta^2 - \frac{\psi_1(x)}{x^2}\right]$, whence

$$\limsup_{x\to\infty} \frac{\psi(x)}{x} \le \frac{1}{\beta-1}\Big[\frac{1}{2}\beta^2 - \frac{1}{2}\Big] = \frac{1}{2}(\beta+1).$$

As $\beta > 1$ is arbitrary, $\limsup \psi(x)/x \le 1$; a similar argument with $\alpha < 1$ gives $\liminf \psi(x)/x \ge 1$. ∎

To relate ψ_1 and ζ: differentiating Lemma 1.3, for $\mathrm{Re}(s) > 1$, $\frac{\zeta'(s)}{\zeta(s)} = -\sum_{m,p}(\log p)p^{-ms} = -\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}$, i.e.

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}. \tag{5}$$

Proposition 2.3

For all $c > 1$,

$$\psi_1(x) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big)\,ds. \tag{6}$$

Lemma 2.4

If $c > 0$, then

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{a^s}{s(s+1)}\,ds = \begin{cases} 0 & \text{if } 0 < a \le 1,\\ 1 - 1/a & \text{if } 1 \le a.\end{cases}$$

Here, the integral is over the vertical line $\mathrm{Re}(s) = c$.

Proof. $|a^s| = a^c$ on the line, so the integral converges. Suppose $a \ge 1$, $a = e^\beta$, $\beta = \log a \ge 0$, and $f(s) = \frac{a^s}{s(s+1)} = \frac{e^{\beta s}}{s(s+1)}$; then $\mathrm{res}_{s=0} f = 1$ and $\mathrm{res}_{s=-1} f = -1/a$. For $T > 0$ let $\Gamma(T)$ be the vertical segment $S(T)$ from $c - iT$ to $c + iT$ together with the half-circle $C(T)$ of radius T centered at c, to the left of the segment, positively oriented (a toy contour). For T large, 0 and -1 are inside, and $\frac{1}{2\pi i}\int_{\Gamma(T)} f = 1 - 1/a$. On $C(T)$, for large T, $|s(s+1)| \ge \frac{1}{2}T^2$ and (as $\sigma \le c$) $|e^{\beta s}| \le e^{\beta c}$, so $\left|\int_{C(T)} f\right| \le \frac{C}{T^2}2\pi T \to 0$. If $0 < a \le 1$, use the half-circle to the right of $\mathrm{Re}(s) = c$; there are no poles inside and the same estimate applies (now $\beta \le 0$ and $\sigma \ge c$). ∎

Proof of Proposition 2.3. $\psi(u) = \sum_{n=1}^{\infty}\Lambda(n)f_n(u)$ with $f_n(u) = 1$ if $n \le u$ and 0 otherwise. Hence

$$\psi_1(x) = \int_0^x \psi(u)\,du = \sum_{n=1}^{\infty}\Lambda(n)\int_0^x f_n(u)\,du = \sum_{n\le x}\Lambda(n)(x-n).$$

By (5) and Lemma 2.4 with $a = x/n$,

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big)ds = x\sum_{n=1}^{\infty}\Lambda(n)\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{(x/n)^s}{s(s+1)}\,ds = x\sum_{n\le x}\Lambda(n)\Big(1-\frac{n}{x}\Big) = \psi_1(x).$$

■

2.1 Proof of the asymptotics for ψ_1

We show $\psi_1(x) \sim x^2/2$, which completes the proof of the prime number theorem. The ingredients: the formula (6); the non-vanishing $\zeta(1+it) \neq 0$; and the estimates of Proposition 2.7 and Proposition 1.6. Strategy: move the line of integration from $\mathrm{Re}(s) = c$ to $\mathrm{Re}(s) = 1$, so that x^{s+1} has size x^2 instead of x^{c+1}; the pole of ζ at $s = 1$ contributes exactly the main term $x^2/2$, and the rest must be shown to be essentially smaller.

Fix $c = 2$, $x \ge 2$, and let

$$F(s) = \frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big).$$

Deform the line $\mathrm{Re}(s) = c$ to the path $\gamma(T)$ ($T \ge 3$, to be chosen): along $\mathrm{Re}(s) = 1$ for $|t| \ge T$, and around $s = 1$ through the segment $\mathrm{Re}(s) = c$, $|t| \le T$, joined by horizontal segments at height $\pm T$. Then

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} F(s)\,ds = \frac{1}{2\pi i}\int_{\gamma(T)} F(s)\,ds. \tag{7}$$

Indeed, by Proposition 2.7 and Proposition 1.6, $|\zeta'(s)/\zeta(s)| \le A_\eta |t|^\eta$ for any fixed $\eta > 0$, when $\sigma \ge 1$, $|t| \ge 1$; so $|F(s)| \le A'|t|^{-2+\eta}$ in the two infinite regions between the line and $\gamma(T)$, where F is regular, and Cauchy's theorem gives (7).

Next pass from $\gamma(T)$ to $\gamma(T,\delta)$, which consists of γ_1: $\mathrm{Re}(s) = 1$, $t \le -T$; γ_2: horizontal segment at $t = -T$ from 1 to $1-\delta$; γ_3: $\mathrm{Re}(s) = 1-\delta$, $|t| \le T$; γ_4: horizontal segment at $t = T$ from $1-\delta$ to 1; γ_5: $\mathrm{Re}(s) = 1$, $t \ge T$. For fixed T, choose $\delta > 0$ so small that ζ has no zeros in the box $\{\sigma + it : 1-\delta \le \sigma \le 1,\ |t| \le T\}$ (possible since ζ does not vanish on $\sigma = 1$). F has a simple pole at $s = 1$: by Corollary 2.6, $\zeta(s) = \frac{1}{s-1} + H(s)$ with H regular near 1, so $-\zeta'(s)/\zeta(s) = \frac{1}{s-1} + h(s)$ with h holomorphic near 1, and $\mathrm{res}_{s=1} F = x^2/2$. Thus

$$\frac{1}{2\pi i}\int_{\gamma(T)} F(s)\,ds = \frac{x^2}{2} + \frac{1}{2\pi i}\int_{\gamma(T,\delta)} F(s)\,ds.$$

Let $\epsilon > 0$ be given. γ_1, γ_5. For $\sigma = 1$, $|x^{1+s}| = x^2$, and by Proposition 1.6 (with Proposition 2.7) $|\zeta'/\zeta| \le A|t|^{1/2}$, so $\big|\int_{\gamma_5} F\big| \le Cx^2\int_T^\infty \frac{|t|^{1/2}}{t^2}\,dt$; the integral converges, so we can choose T with $\big|\int_{\gamma_1} F\big|, \big|\int_{\gamma_5} F\big| \le \epsilon x^2$. γ_3. With T fixed, $|x^{1+s}| = x^{2-\delta}$ on γ_3, so $\big|\int_{\gamma_3} F\big| \le C_T x^{2-\delta}$. γ_2, γ_4. $\big|\int_{\gamma_2} F\big| \le C'_T\int_{1-\delta}^1 x^{1+\sigma}\,d\sigma \le C'_T\frac{x^2}{\log x}$.

Hence there are constants C_T, C'_T with

$$\Big|\psi_1(x) - \frac{x^2}{2}\Big| \le \epsilon x^2 + C_T x^{2-\delta} + C'_T\frac{x^2}{\log x}, \qquad \Big|\frac{2\psi_1(x)}{x^2} - 1\Big| \le 2\epsilon + 2C_T x^{-\delta} + \frac{2C'_T}{\log x},$$

and for all large x, $\big|\frac{2\psi_1(x)}{x^2} - 1\big| \le 4\epsilon$. So $\psi_1(x) \sim x^2/2$, and the prime number theorem is proved.

Note on interchanging double sums

Proposition

Let $\{a_{k\ell}\}_{1\le k,\ell<\infty}$ be complex numbers with

$$\sum_{k=1}^{\infty}\Big(\sum_{\ell=1}^{\infty}|a_{k\ell}|\Big)<\infty. \tag{8}$$

Then:
(i) The double sum $A=\sum_{k=1}^{\infty}\sum_{\ell=1}^{\infty}a_{k\ell}$ converges, and the order of summation may be interchanged: $A=\sum_{\ell=1}^{\infty}\sum_{k=1}^{\infty}a_{k\ell}$.
(ii) Given $\epsilon>0$, there is N so that for all $K,L>N$, $|A-\sum_{k=1}^{K}\sum_{\ell=1}^{L}a_{k\ell}|<\epsilon$.
(iii) If $m\mapsto(k(m),\ell(m))$ is a bijection from $\mathbb{N}$ to $\mathbb{N}\times\mathbb{N}$ and $c_m=a_{k(m)\ell(m)}$, then $A=\sum_{m=1}^{\infty}c_m$.

Proof. (i) Let $b_k=\sum_\ell|a_{k\ell}|$, and let $S=\{x_0,x_1,\dots\}$ with $x_n\to x_0$. On S define $f_k(x_0)=\sum_{\ell=1}^{\infty}a_{k\ell}$, $f_k(x_n)=\sum_{\ell=1}^{n}a_{k\ell}$, and $g(x)=\sum_k f_k(x)$. Each f_k is continuous at x_0; $|f_k(x)|\le b_k$ with $\sum b_k<\infty$, so the series for g converges uniformly on S and g is continuous at x_0. Hence

$$\sum_k\sum_\ell a_{k\ell}=g(x_0)=\lim_n g(x_n)=\lim_n\sum_k\sum_{\ell=1}^{n}a_{k\ell}=\lim_n\sum_{\ell=1}^{n}\sum_k a_{k\ell}=\sum_\ell\sum_k a_{k\ell}.$$

(ii) $|A-\sum_{k\le K}\sum_{\ell\le L}a_{k\ell}|\le\sum_{k\le K}\sum_{\ell>L}|a_{k\ell}|+\sum_{k>K}\sum_\ell|a_{k\ell}|$. Choose N with $\sum_{k>N}b_k<\epsilon/2$; the first sum is $\le\sum_{k\le N}\sum_{\ell>L}|a_{k\ell}|+\sum_{k>N}b_k$, and for L large the finitely many tails $\sum_{\ell>L}|a_{k\ell}|$, $k\le N$, are small. [Supplement]

(iii) [Supplement]Given ϵ, take N as in (ii) (for the absolute values as well), and M so large that $\{c_1,\dots,c_M\}$ contains all $a_{k\ell}$ with $k,\ell\le N$. For $M'\ge M$, $\sum_{m\le M'}c_m-\sum_{k,\ell\le N}a_{k\ell}$ is a sum of terms $a_{k\ell}$ with $k>N$ or $\ell>N$, of total absolute value $<\epsilon$; so $|\sum_{m\le M'}c_m-A|<2\epsilon$. ∎

3 Selected exercises

Motivation. Four exercises from Stein–Shakarchi, Chapter 7 (book numbers kept): the Dirichlet series of $1/\zeta$, the alternating zeta function and the absence of real zeros in the critical strip, the reality of ξ, and the size of the n-th prime.

Exercise 7.3 (The Möbius function)

Let $\mu(1)=1$, $\mu(n)=(-1)^k$ if $n=p_1\cdots p_k$ with distinct primes p_j, and $\mu(n)=0$ otherwise.
(a) Prove that $\dfrac{1}{\zeta(s)}=\sum_{n=1}^{\infty}\dfrac{\mu(n)}{n^s}$ for $\mathrm{Re}(s)>1$.
(b) Show that $\sum_{k|n}\mu(k)$ equals 1 if $n=1$ and 0 otherwise.

Solution.

(a) Let $\mathrm{Re}(s)=\sigma>1$.

Step 1. By Euler's identity the partial products $\prod_{p\le N}(1-p^{-s})^{-1}$ tend to $\zeta(s)\neq0$, so their reciprocals tend to $1/\zeta(s)$:

$$\frac{1}{\zeta(s)}=\lim_{N\to\infty}\prod_{p\le N}(1-p^{-s}).$$

Step 2. Expanding the finite product, each term corresponds to a set S of primes $\le N$ and equals $\prod_{p\in S}(-p^{-s})=\mu(n)n^{-s}$ with $n=\prod_{p\in S}p$. Every square-free n whose prime factors are all $\le N$

arises exactly once, and $\mu(n) = 0$ for the other n. Hence

$$\prod_{p \le N} (1 - p^{-s}) = \sum_{n \in \mathcal{A}_N} \frac{\mu(n)}{n^s}, \qquad \mathcal{A}_N = \{n : \text{all prime factors of } n \text{ are} \le N\}.$$

Step 3. Every $n \le N$ lies in $\mathcal{A}_N$ and $|\mu(n)| \le 1$. So

$$\Big| \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s} - \sum_{n \in \mathcal{A}_N} \frac{\mu(n)}{n^s} \Big| \le \sum_{n > N} n^{-\sigma} \to 0.$$

Combining with Step 1 gives (a).

(b) For $n = 1$ the sum is $\mu(1) = 1$. Let $n = p_1^{a_1} \cdots p_r^{a_r}$ with $r \ge 1$ and all $a_j \ge 1$. The divisors k with $\mu(k) \ne 0$ are the products $\prod_{j \in S} p_j$ over subsets $S \subseteq \{1, \dots, r\}$, and for these $\mu(k) = (-1)^{|S|}$. Hence

$$\sum_{k \mid n} \mu(k) = \sum_{S \subseteq \{1,\dots,r\}} (-1)^{|S|} = \sum_{j=0}^{r} \binom{r}{j} (-1)^j = (1-1)^r = 0.$$

■

Exercise 7.5 (The alternating zeta function)

Let $\tilde{\zeta}(s) = \sum_{n=1}^{\infty} (-1)^{n+1} n^{-s} = 1 - 2^{-s} + 3^{-s} - \cdots$.
(a) Prove that the series converges for $\mathrm{Re}(s) > 0$ and defines a holomorphic function there.
(b) Show that $\tilde{\zeta}(s) = (1 - 2^{1-s})\zeta(s)$ for $s > 1$.
(c) Conclude that ζ has no zeros on the segment $0 < \sigma < 1$ of the real axis, and that $\zeta(0) \ne 0$.

Solution.

(a) Let K be a compact subset of $\{\mathrm{Re}(s) > 0\}$, with $|s| \le M$ and $\mathrm{Re}(s) \ge \delta > 0$ on K.

Step 1. Group the terms in pairs. Since $(2k-1)^{-s} - (2k)^{-s} = s \int_{2k-1}^{2k} x^{-s-1}\, dx$,

$$|(2k-1)^{-s} - (2k)^{-s}| \le |s|\,(2k-1)^{-\delta-1} \le M(2k-1)^{-1-\delta} \qquad (s \in K).$$

So the even partial sums $S_{2N}(s) = \sum_{k \le N} [(2k-1)^{-s} - (2k)^{-s}]$ converge uniformly on K.

Step 2. The odd partial sums satisfy $S_{2N+1} - S_{2N} = (2N+1)^{-s}$, which has modulus $\le (2N+1)^{-\delta} \to 0$ uniformly on K. Hence all partial sums converge uniformly on K.

Step 3. The partial sums are entire, so $\tilde{\zeta}$ is holomorphic in $\mathrm{Re}(s) > 0$ by Theorem 5.2.

(b) For $s > 1$ all series converge absolutely and may be rearranged:

$$\zeta(s) - \tilde{\zeta}(s) = 2 \sum_{m \ge 1} (2m)^{-s} = 2^{1-s}\zeta(s).$$

(c)

Step 1. Both $\tilde{\zeta}$ and $(1 - 2^{1-s})\zeta(s)$ are holomorphic on the connected set $\{\mathrm{Re}(s) > 0\} \setminus \{1\}$ (Corollary 2.6). They agree on $(1, \infty)$ by (b), so they agree on that whole set (Theorem 4.8).

Step 2. For real $0 < \sigma < 1$, $x \mapsto x^{-\sigma}$ is strictly decreasing, so every bracket in $S_{2N}(\sigma)$ is positive and increasing in N. Hence $\tilde{\zeta}(\sigma) = \lim S_{2N}(\sigma) \ge 1 - 2^{-\sigma} > 0$. Also $1 - 2^{1-\sigma} < 0$. So $\zeta(\sigma) = \tilde{\zeta}(\sigma)/(1 - 2^{1-\sigma}) < 0$; in particular $\zeta(\sigma) \ne 0$.

Step 3. ($s = 0$.) The proof of Theorem 2.3 gives $\xi(s) = \frac{1}{s-1} - \frac{1}{s} + E(s)$ with E entire, so $s\,\xi(s) \to -1$ as $s \to 0$. Also $\frac{s}{2}\Gamma(\frac{s}{2}) = \Gamma(\frac{s}{2} + 1) \to 1$. Writing

$$\zeta(s) = \pi^{s/2} \frac{\xi(s)}{\Gamma(s/2)} = \pi^{s/2} \cdot s\,\xi(s) \cdot \frac{1}{2 \cdot \frac{s}{2}\Gamma(\frac{s}{2})},$$

we get $\zeta(0) = 1 \cdot (-1) \cdot \frac{1}{2} = -\frac{1}{2} \ne 0$. ■

Exercise 7.7 (Reality of ξ)

Show that $\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ is real when s is real ($s \neq 0, 1$) and when $\mathrm{Re}(s) = 1/2$.

Solution.

Step 1. For real $s > 1$, each of $\pi^{-s/2}$, $\Gamma(s/2) = \int_0^\infty e^{-u}u^{s/2-1}du$ and $\zeta(s) = \sum n^{-s}$ is real, so $\xi(s) \in \mathbb{R}$.

Step 2. Let $g(s) = \overline{\xi(\overline{s})}$. As in the Schwarz reflection principle (Theorem 5.6), g is holomorphic on $\Omega = \mathbb{C} \setminus \{0, 1\}$, since its power series at s_0 has the conjugate coefficients of those of ξ at $\overline{s_0}$. By Step 1, $g = \xi$ on $(1, \infty)$. As Ω is connected, Theorem 4.8 gives $\xi(\overline{s}) = \overline{\xi(s)}$ on Ω.

Step 3. If $s \in \mathbb{R} \setminus \{0, 1\}$, Step 2 gives $\xi(s) = \overline{\xi(s)}$, so $\xi(s)$ is real.

Step 4. If $\mathrm{Re}(s) = 1/2$, then $\overline{s} = 1 - s$. By Theorem 2.3, $\overline{\xi(s)} = \xi(\overline{s}) = \xi(1 - s) = \xi(s)$, so $\xi(s)$ is real. ■

Exercise 7.12 (The n-th prime)

Let p_n denote the n-th prime. Use the prime number theorem to show that $p_n \sim n \log n$ as $n \to \infty$.

Solution.

Step 1. For $x \geq 3$ we have $\pi(x) \geq 1$ and $\log\log x > 0$. By the prime number theorem $\pi(x)\log x/x \to 1$, so

$$\log \pi(x) + \log\log x - \log x = \log \frac{\pi(x)\log x}{x} \to 0.$$

Step 2. Dividing by $\log x \to \infty$, and using $\log\log x/\log x \to 0$,

$$\frac{\log \pi(x)}{\log x} = \frac{\log \pi(x) + \log\log x - \log x}{\log x} + 1 - \frac{\log\log x}{\log x} \to 1.$$

So $\log \pi(x) \sim \log x$.

Step 3. Multiplying the two asymptotic relations,

$$\frac{\pi(x)\log \pi(x)}{x} = \frac{\pi(x)\log x}{x} \cdot \frac{\log \pi(x)}{\log x} \to 1.$$

Step 4. Take $x = p_n \to \infty$. Then $\pi(p_n) = n$, so $n \log n/p_n \to 1$, i.e. $p_n \sim n \log n$. ■